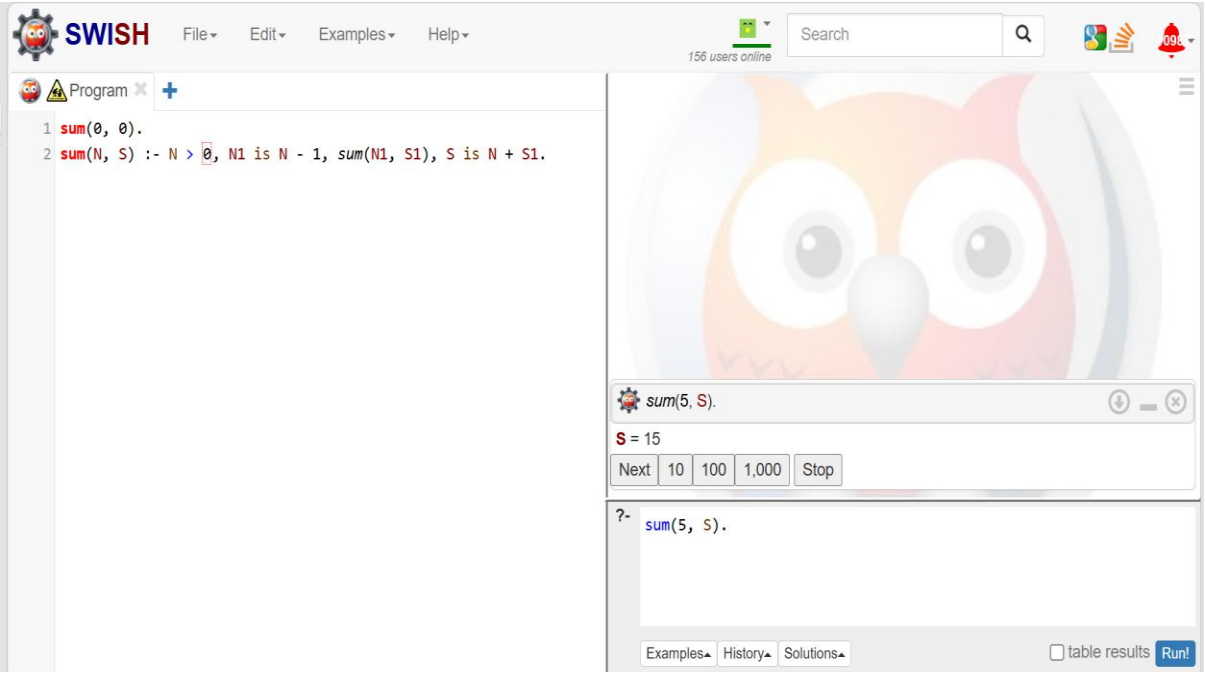
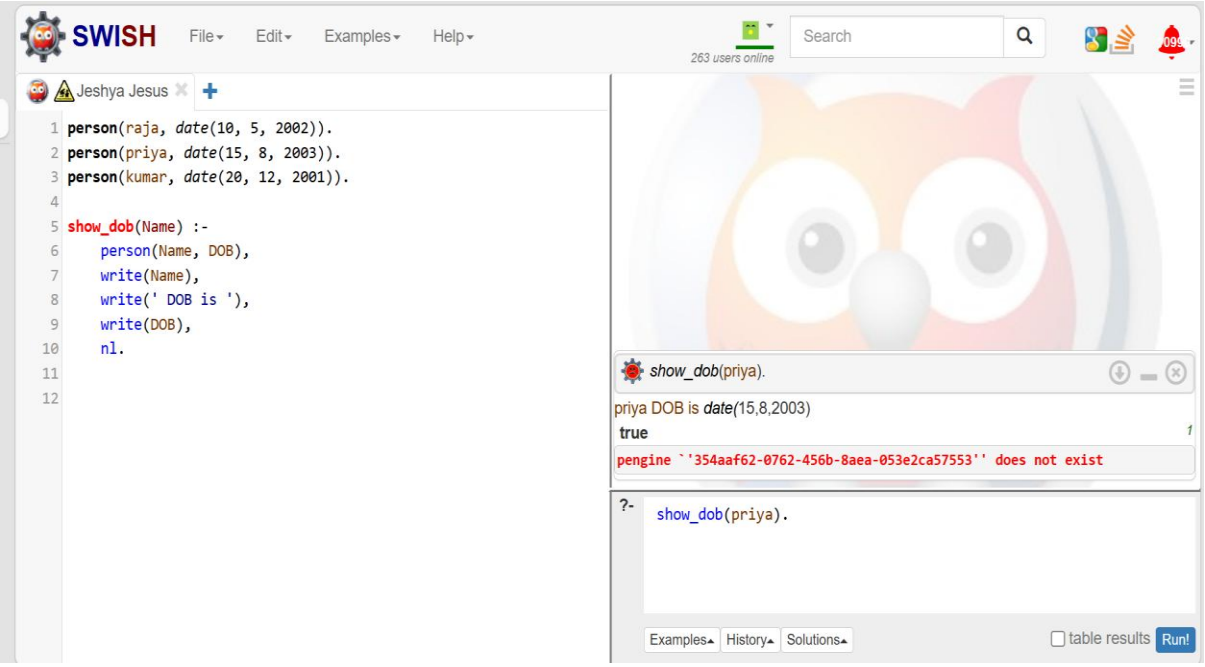
PROLOG EXPERIMENTS

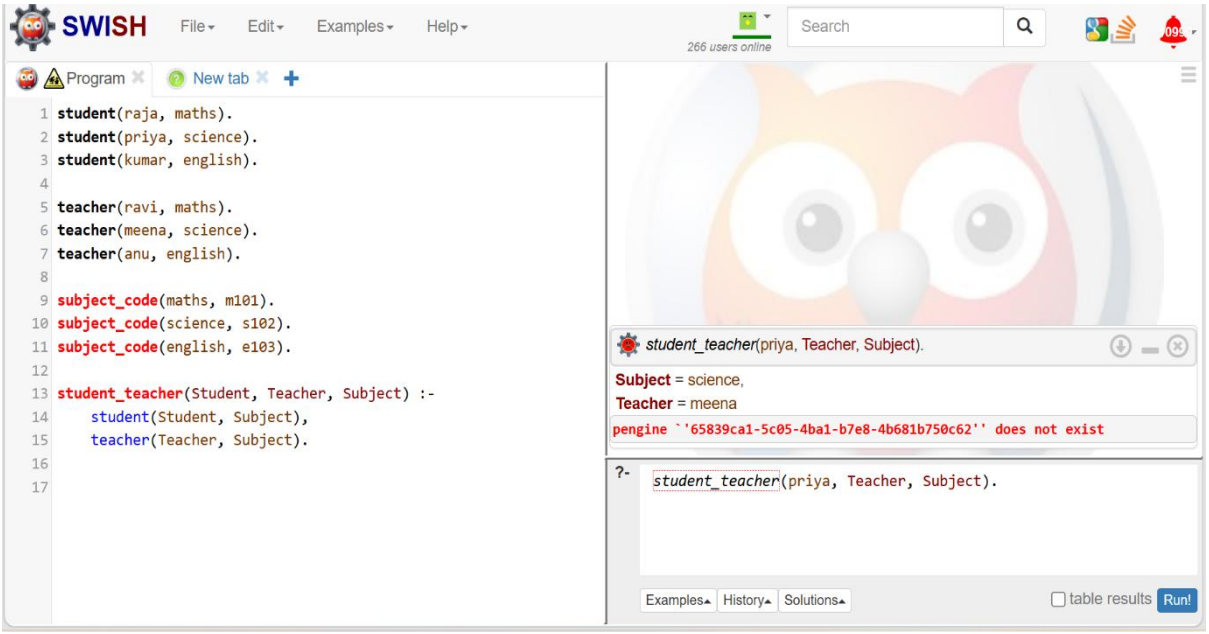
EX:17



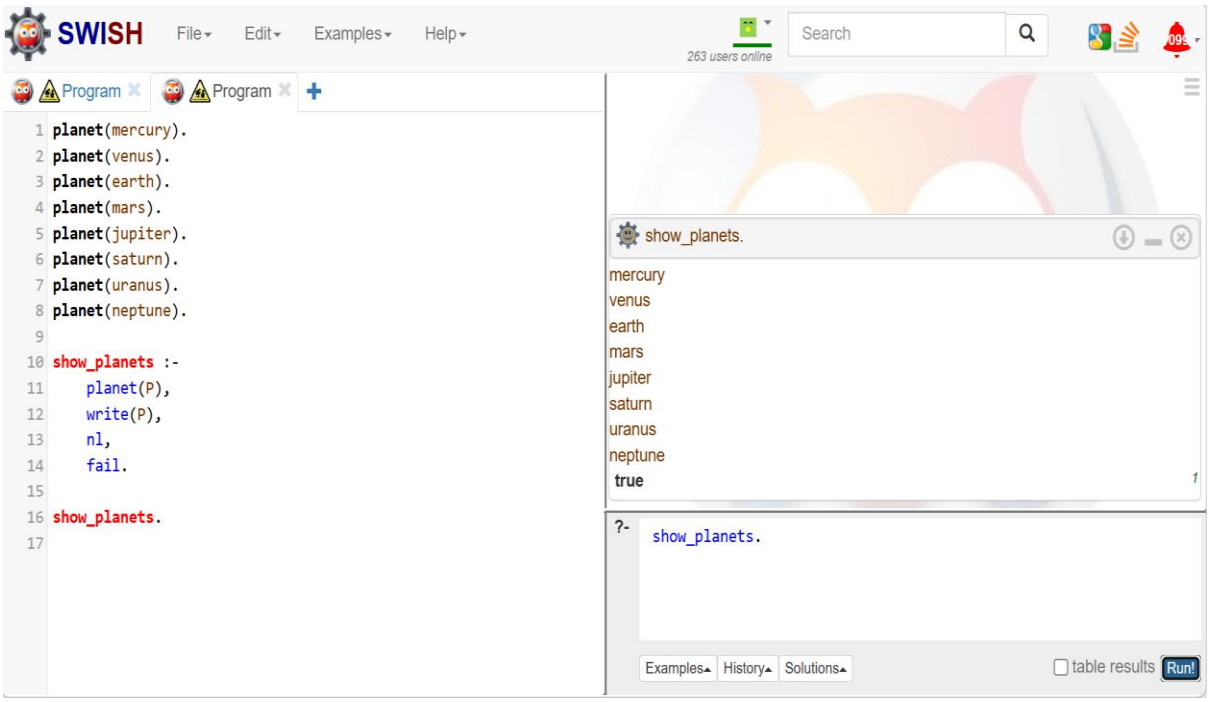
EX:18



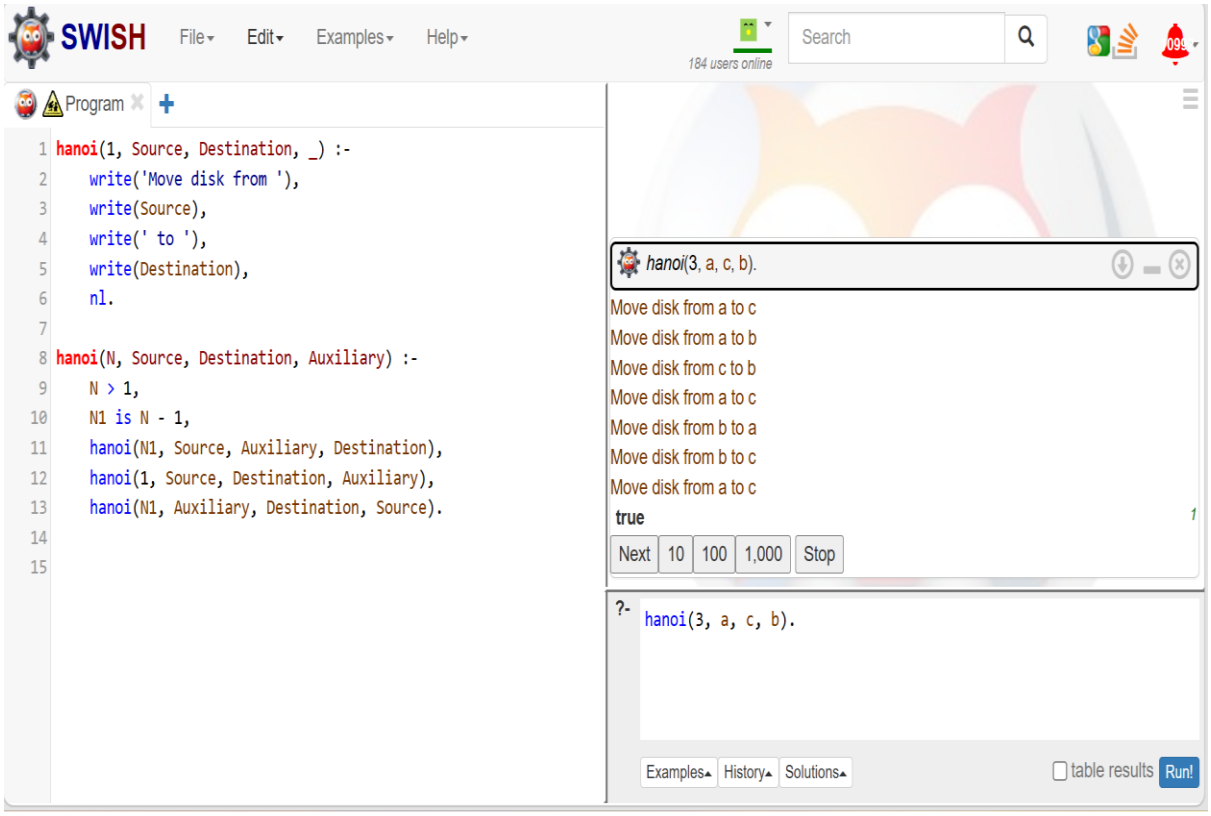
EX:19


SWISH
File
Edit
Examples
Help
266 users online
Search
Program
New tab
1 student(raja, maths).
2 student(priya, science).
3 student(kumar, english).
4
5 teacher(ravi, maths).
6 teacher(meena, science).
7 teacher(anu, english).
8
9 subject_code(maths, m101).
10 subject_code(science, s102).
11 subject_code(english, e103).
12
13 student_teacher(Student, Teacher, Subject) :-
14 student(Student, Subject),
15 teacher(Teacher, Subject).
16
17
student_teacher(priya, Teacher, Subject).
Subject = science,
Teacher = meena
pengine ``'65839ca1-5c05-4ba1-b7e8-4b681b750c62'' does not exist
?- student_teacher(priya, Teacher, Subject).
Examples
History
Solutions
table results
Run!

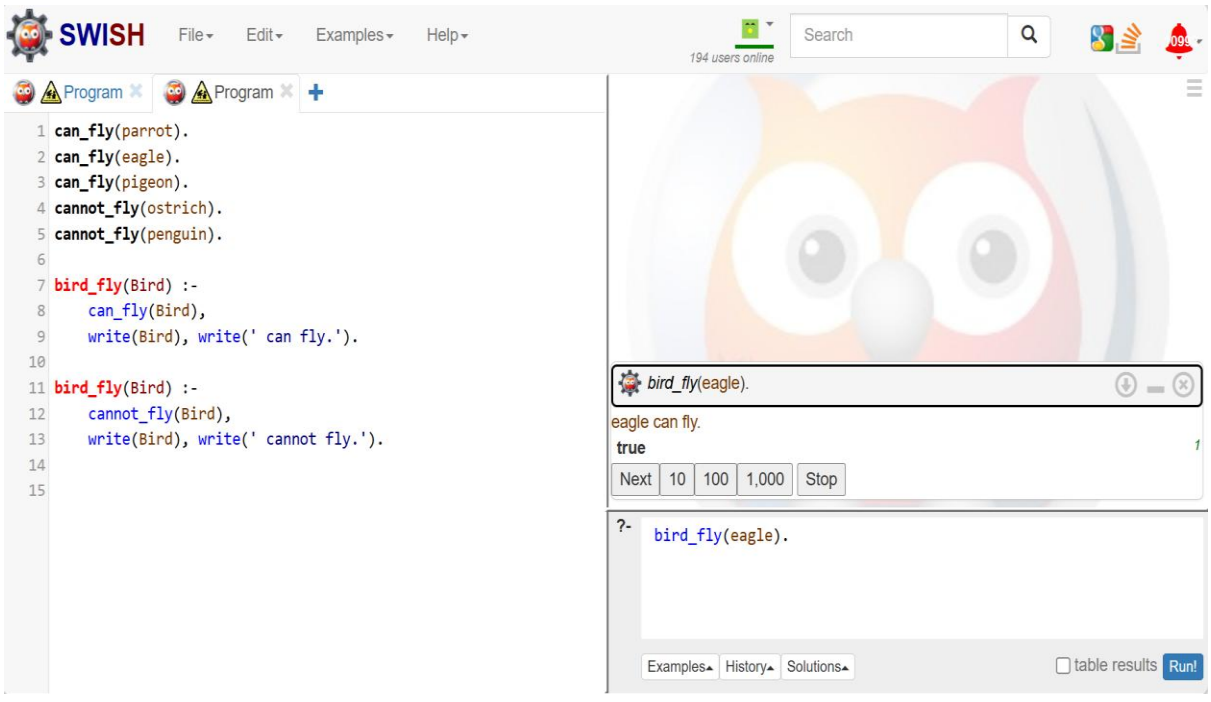
EX:20


SWISH
File
Edit
Examples
Help
263 users online
Search
Program
Program
1 planet(mercury).
2 planet(venus).
3 planet(earth).
4 planet(mars).
5 planet(jupiter).
6 planet(saturn).
7 planet(uranus).
8 planet(neptune).
9
10 show_planets :-
11 planet(P),
12 write(P),
13 nl,
14 fail.
15
16 show_planets.
17
show_planets.
mercury
venus
earth
mars
jupiter
saturn
uranus
neptune
true
?- show_planets.
Examples
History
Solutions
table results
Run!

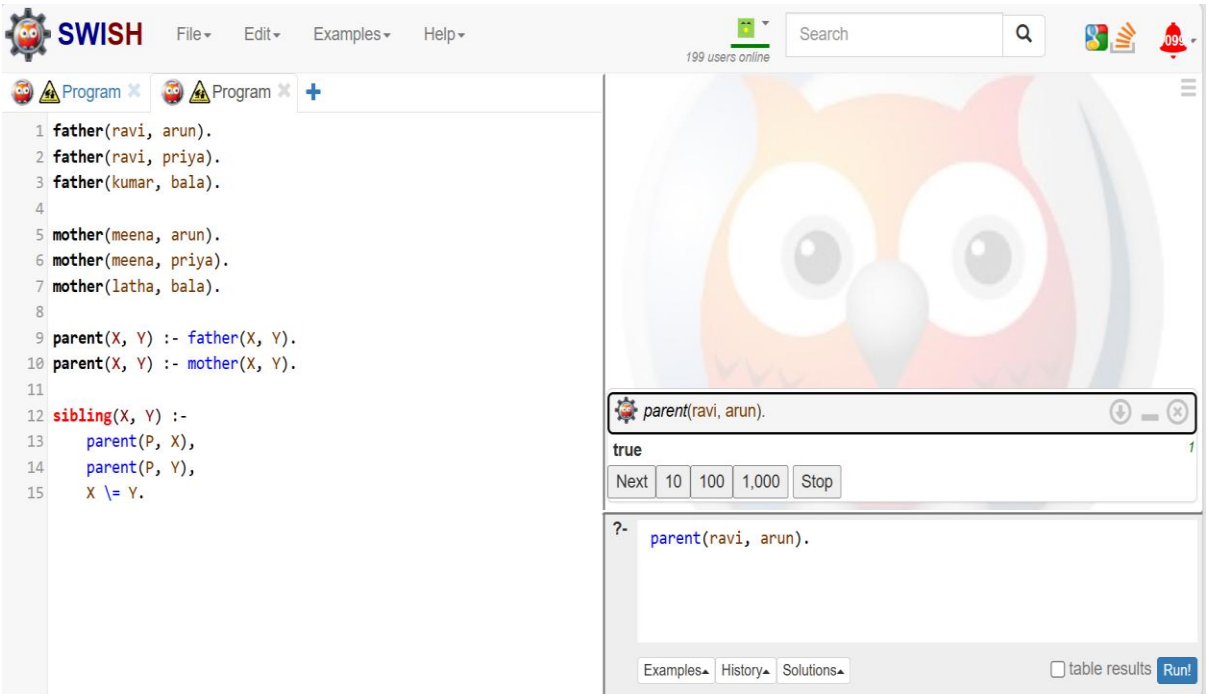
EX:21


SWISH
File
Edit
Examples
Help
184 users online
Search
Program
hanoi(1, Source, Destination, _) :-
write('Move disk from '),
write(Source),
write(' to '),
write(Destination),
nl.
hanoi(N, Source, Destination, Auxiliary) :-
N > 1,
N1 is N - 1,
hanoi(N1, Source, Auxiliary, Destination),
hanoi(1, Source, Destination, Auxiliary),
hanoi(N1, Auxiliary, Destination, Source).
hanoi(3, a, c, b).
Move disk from a to c
Move disk from a to b
Move disk from c to b
Move disk from a to c
Move disk from b to a
Move disk from b to c
Move disk from a to c
true
Next
10
100
1,000
Stop
?-
hanoi(3, a, c, b).
Examples
History
Solutions
table results
Run!

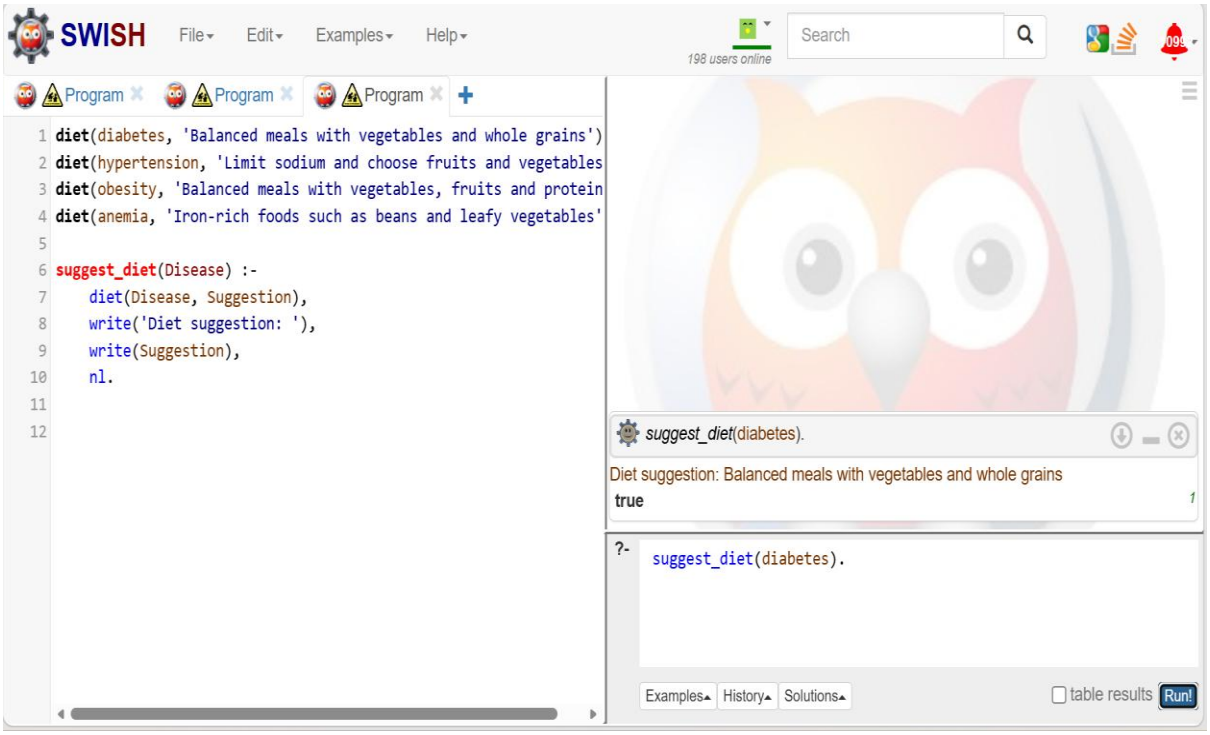
EX:22


SWISH
File
Edit
Examples
Help
194 users online
Search
Program
Program
can_fly(parrot).
can_fly(eagle).
can_fly(pigeon).
cannot_fly(ostrich).
cannot_fly(penguin).
bird_fly(Bird) :-
can_fly(Bird),
write(Bird), write(' can fly.').
bird_fly(Bird) :-
cannot_fly(Bird),
write(Bird), write(' cannot fly.').
bird_fly(eagle).
eagle can fly.
true
Next
10
100
1,000
Stop
?-
bird_fly(eagle).
Examples
History
Solutions
table results
Run!

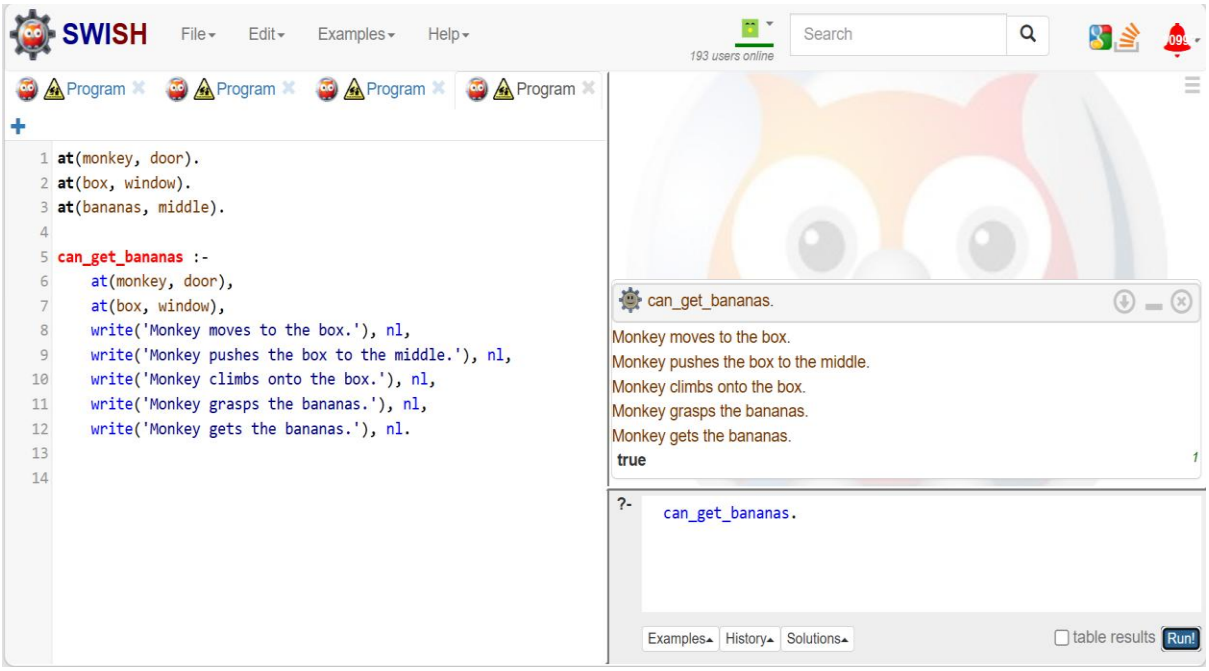
EX:23


SWISH
File
Edit
Examples
Help
199 users online
Search
Program
Program
father(ravi, arun).
father(ravi, priya).
father(kumar, bala).
mother(meena, arun).
mother(meena, priya).
mother(latha, bala).
parent(X, Y) :- father(X, Y).
parent(X, Y) :- mother(X, Y).
sibling(X, Y) :-
parent(P, X),
parent(P, Y),
X \= Y.
parent(ravi, arun).
true
Next
10
100
1,000
Stop
?-
parent(ravi, arun).
Examples
History
Solutions
table results
Run!

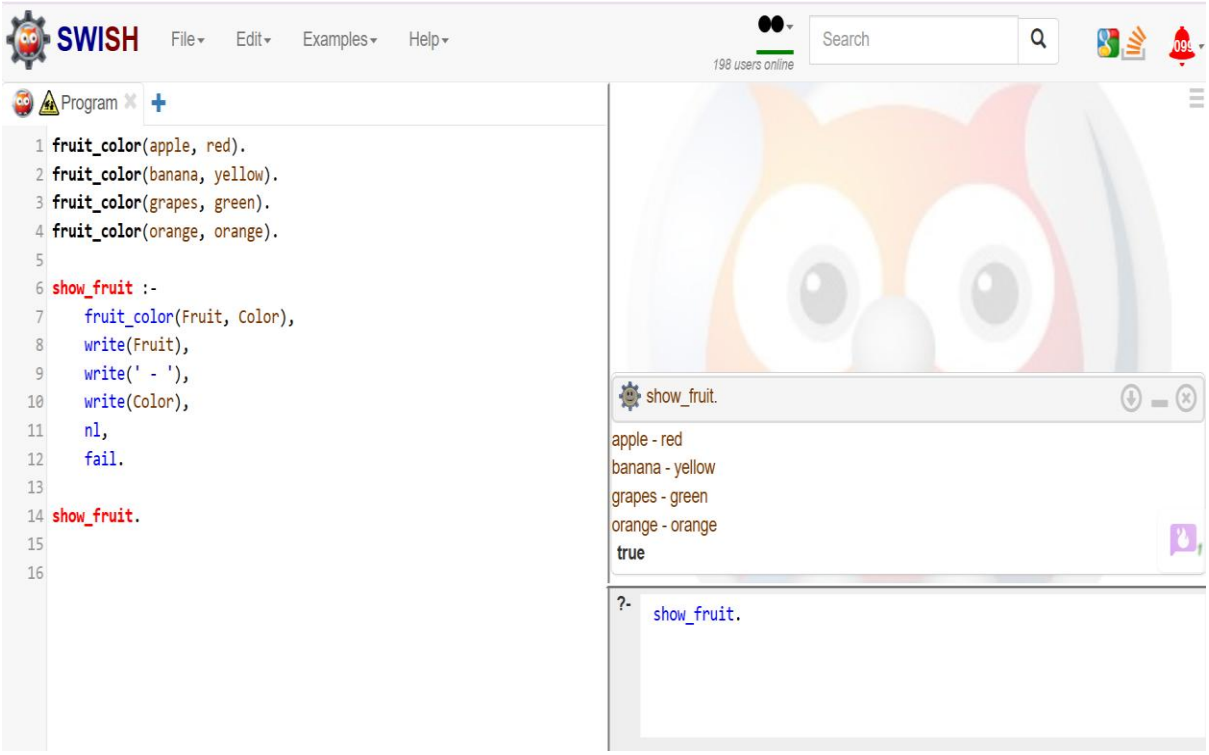
EX:24


SWISH
File
Edit
Examples
Help
198 users online
Search
Program
Program
Program
diet(diabetes, 'Balanced meals with vegetables and whole grains')
diet(hypertension, 'Limit sodium and choose fruits and vegetables
diet(obesity, 'Balanced meals with vegetables, fruits and protein
diet(anemia, 'Iron-rich foods such as beans and leafy vegetables'
suggest_diet(Disease) :-
diet(Disease, Suggestion),
write('Diet suggestion: '),
write(Suggestion),
nl.
suggest_diet(diabetes).
Diet suggestion: Balanced meals with vegetables and whole grains
true
?-
suggest_diet(diabetes).
Examples
History
Solutions
table results
Run!

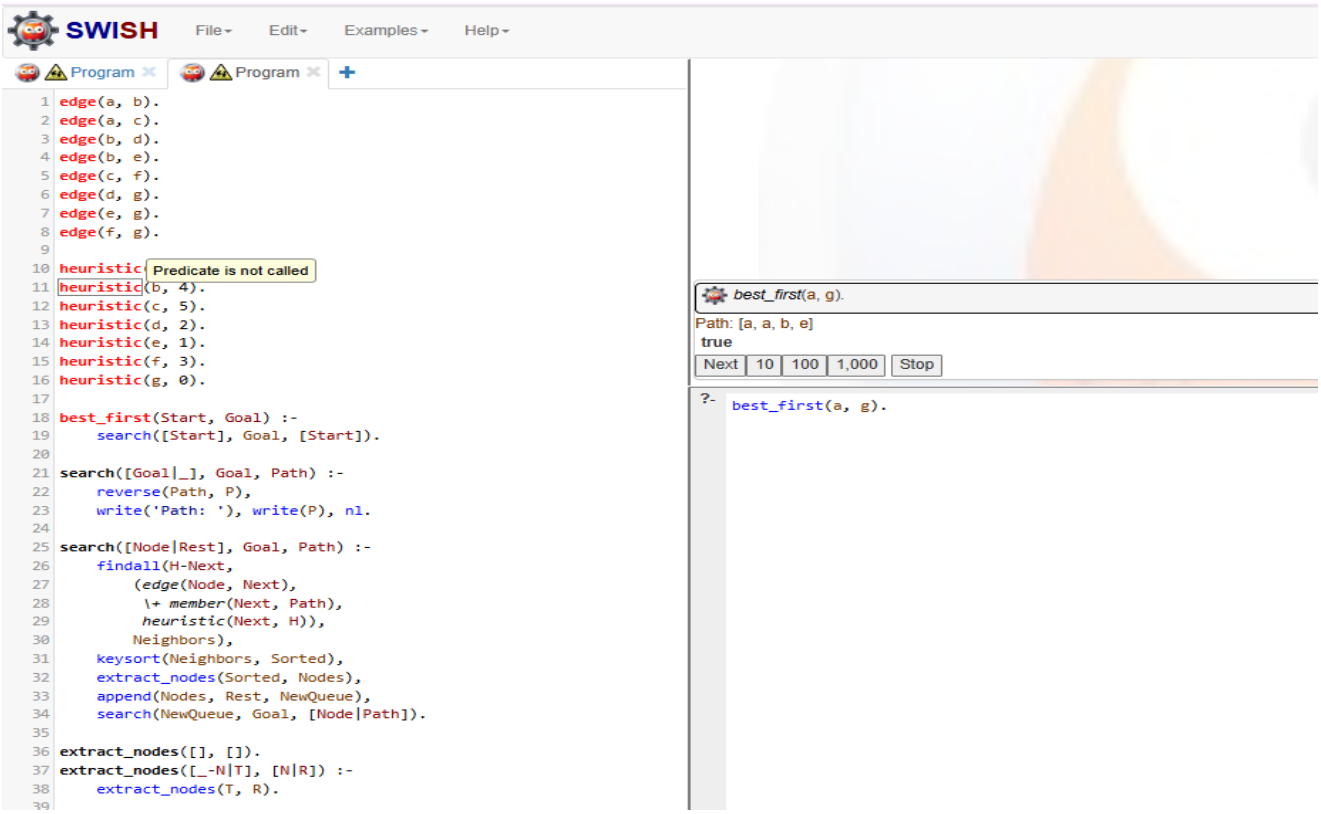
EX:25


SWISH
File
Edit
Examples
Help
193 users online
Search
Program
Program
Program
Program
1 at(monkey, door).
2 at(box, window).
3 at(bananas, middle).
4
5 can_get_bananas :-
6 at(monkey, door),
7 at(box, window),
8 write('Monkey moves to the box.'), nl,
9 write('Monkey pushes the box to the middle.'), nl,
10 write('Monkey climbs onto the box.'), nl,
11 write('Monkey grasps the bananas.'), nl,
12 write('Monkey gets the bananas.'), nl.
13
14
can_get_bananas.
Monkey moves to the box.
Monkey pushes the box to the middle.
Monkey climbs onto the box.
Monkey grasps the bananas.
Monkey gets the bananas.
true
1
?-
can_get_bananas.
Examples
History
Solutions
table results
Run!

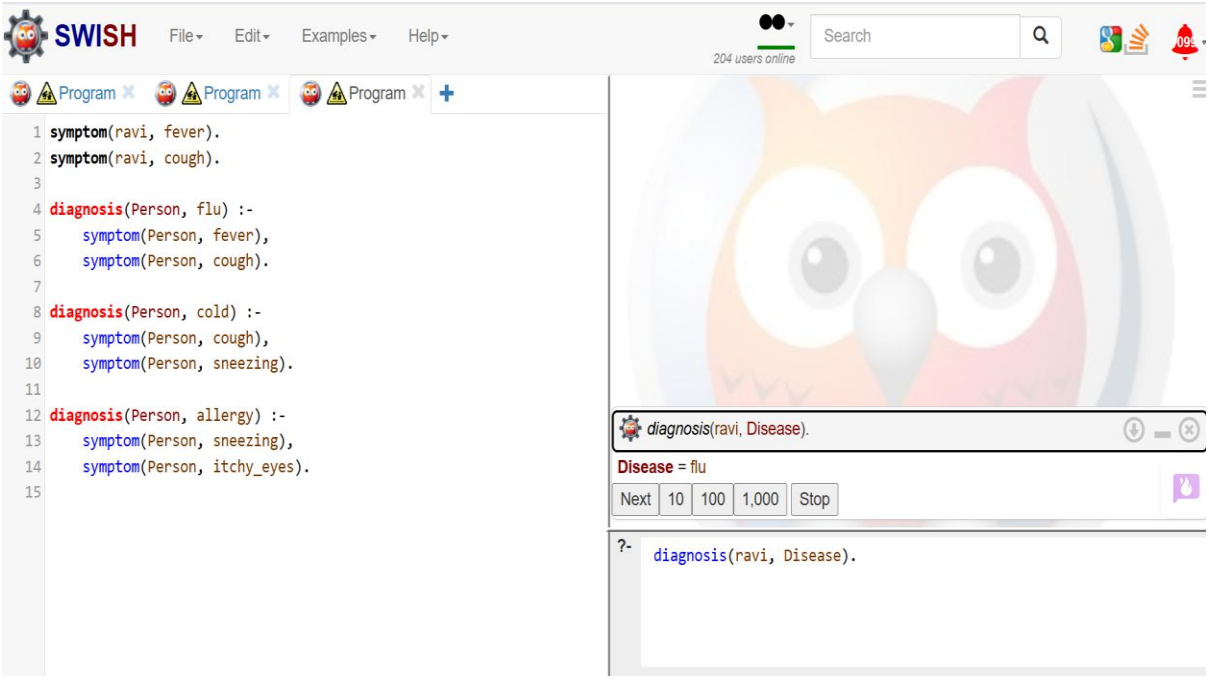
EX:26


SWISH
File
Edit
Examples
Help
198 users online
Search
Program
1 fruit_color(apple, red).
2 fruit_color(banana, yellow).
3 fruit_color(grapes, green).
4 fruit_color(orange, orange).
5
6 show_fruit :-
7 fruit_color(Fruit, Color),
8 write(Fruit),
9 write(' - '),
10 write(Color),
11 nl,
12 fail.
13
14 show_fruit.
15
16
show_fruit.
apple - red
banana - yellow
grapes - green
orange - orange
true
?-
show_fruit.

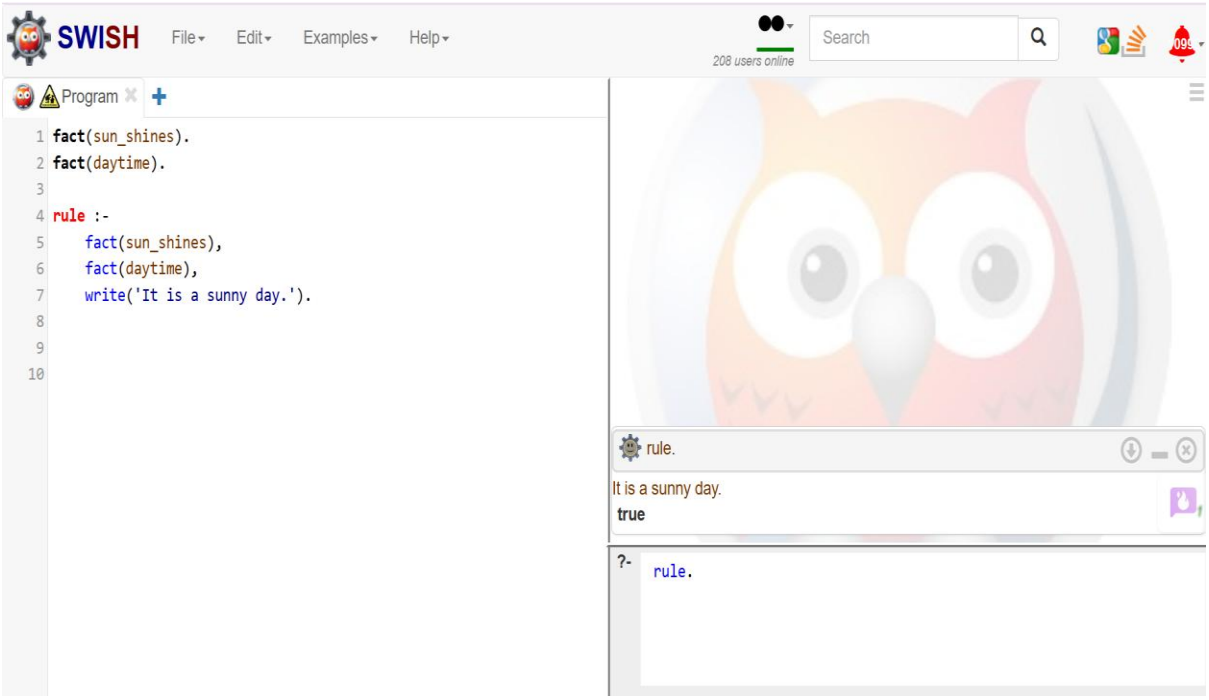
EX:27


SWISH
File
Edit
Examples
Help
Program
Program
edge(a, b).
edge(a, c).
edge(b, d).
edge(b, e).
edge(c, f).
edge(d, g).
edge(e, g).
edge(f, g).
heuristic
Predicate is not called
heuristic(b, 4).
heuristic(c, 5).
heuristic(d, 2).
heuristic(e, 1).
heuristic(f, 3).
heuristic(g, 0).
best_first(Start, Goal) :-
search([Start], Goal, [Start]).
search([Goal|_], Goal, Path) :-
reverse(Path, P),
write('Path: '), write(P), nl.
search([Node|Rest], Goal, Path) :-
findall(H-Next,
(edge(Node, Next),
\+ member(Next, Path),
heuristic(Next, H)),
Neighbors),
keysort(Neighbors, Sorted),
extract_nodes(Sorted, Nodes),
append(Nodes, Rest, NewQueue),
search(NewQueue, Goal, [Node|Path]).
extract_nodes([], []).
extract_nodes([_-N|T], [N|R]) :-
extract_nodes(T, R).
best_first(a, g).
Path: [a, a, b, e]
true
Next
10
100
1,000
Stop
?-
best_first(a, g).

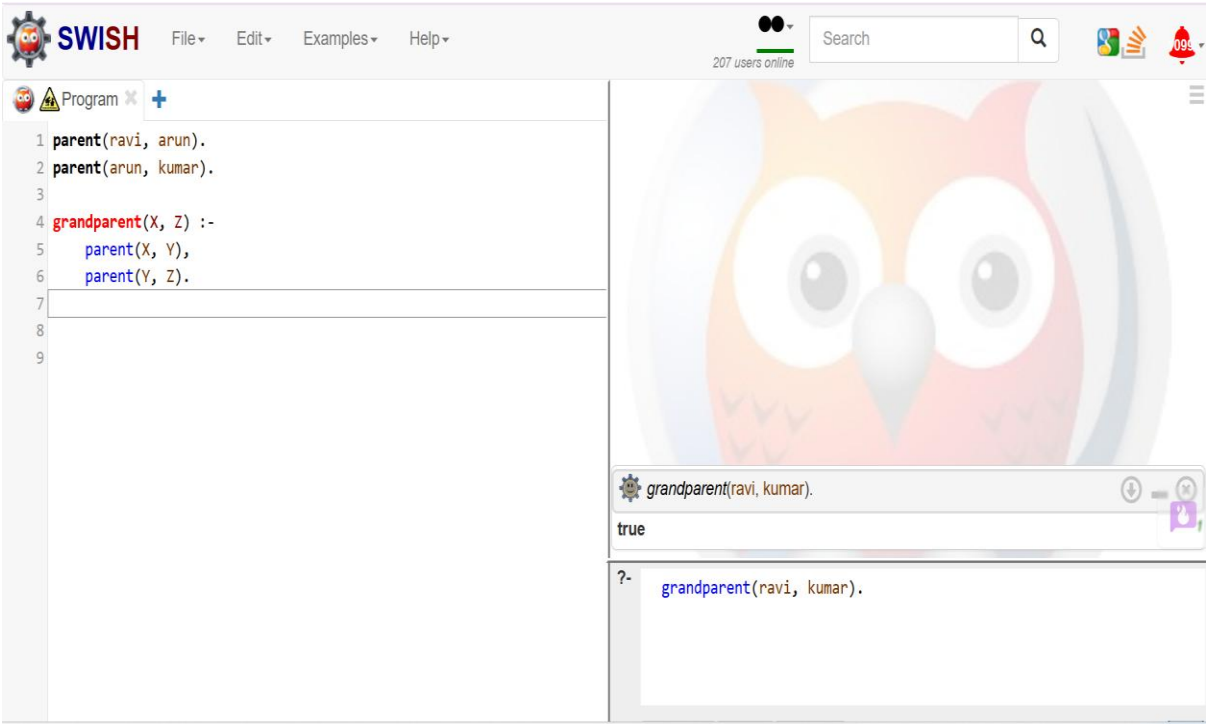
EX:28


SWISH
File
Edit
Examples
Help
Search
204 users online
Program
Program
Program
symptom(ravi, fever).
symptom(ravi, cough).
diagnosis(Person, flu) :-
symptom(Person, fever),
symptom(Person, cough).
diagnosis(Person, cold) :-
symptom(Person, cough),
symptom(Person, sneezing).
diagnosis(Person, allergy) :-
symptom(Person, sneezing),
symptom(Person, itchy_eyes).
diagnosis(ravi, Disease).
Disease = flu
Next
10
100
1,000
Stop
?-
diagnosis(ravi, Disease).

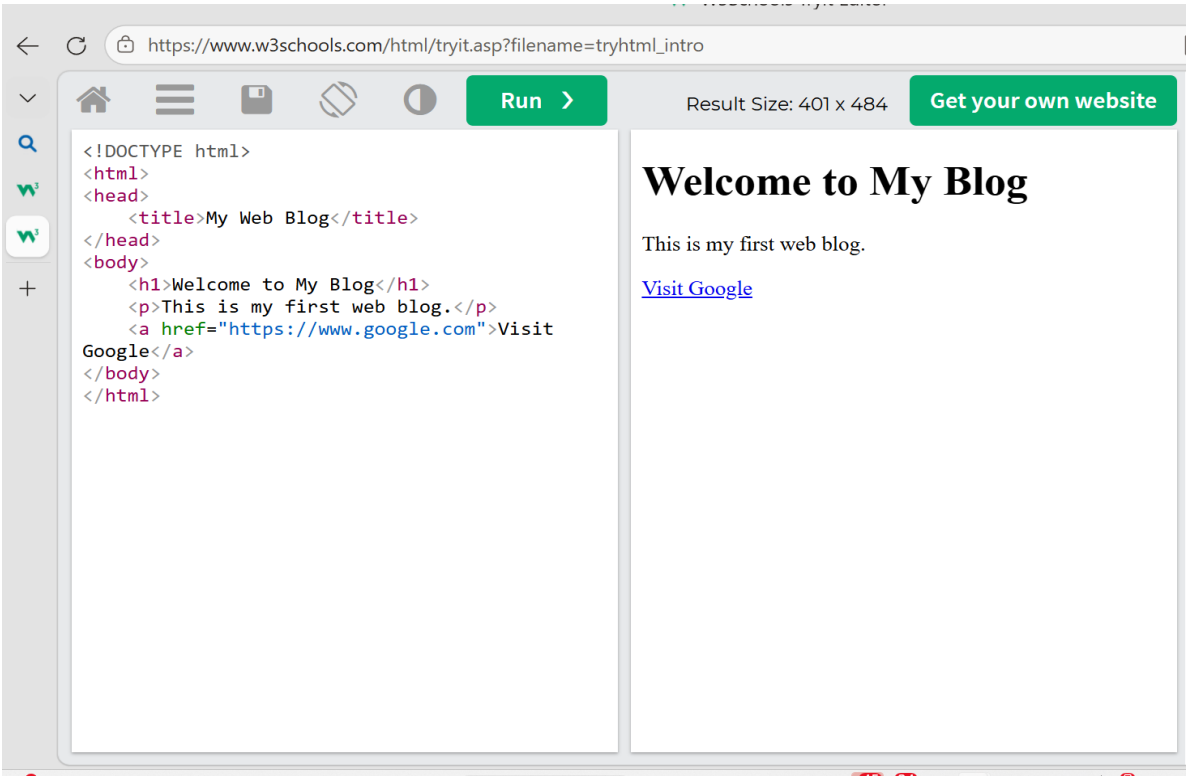
EX:29


SWISH
File
Edit
Examples
Help
208 users online
Search
Program
1 fact(sun_shines).
2 fact(daytime).
3
4 rule :-
5 fact(sun_shines),
6 fact(daytime),
7 write('It is a sunny day.').
8
9
10
rule.
It is a sunny day.
true
?-
rule.

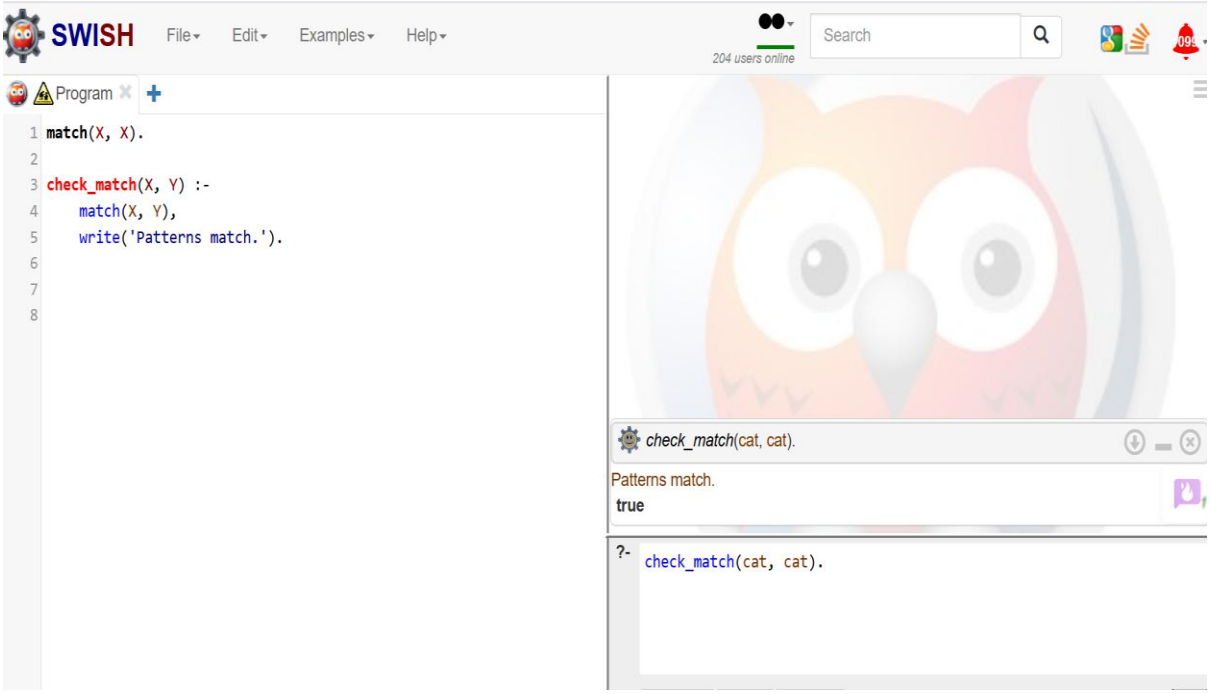
EX:30


SWISH
File
Edit
Examples
Help
207 users online
Search
Program
1 parent(ravi, arun).
2 parent(arun, kumar).
3
4 grandparent(X, Z) :-
5 parent(X, Y),
6 parent(Y, Z).
7
8
9
grandparent(ravi, kumar).
true
?-
grandparent(ravi, kumar).

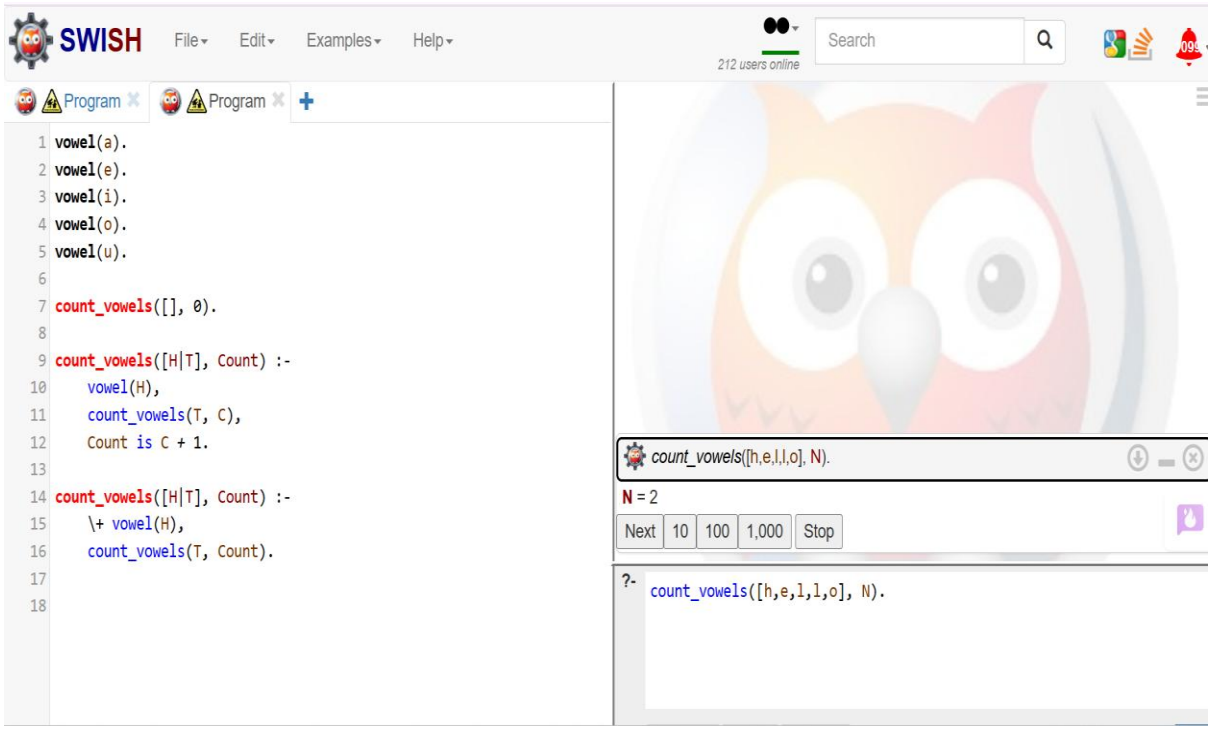
EX:31


https://www.w3schools.com/html/tryit.asp?filename=tryhtml_intro
Run
Result Size: 401 x 484
Get your own website
<!DOCTYPE html>
<html>
<head>
<title>My Web Blog</title>
</head>
<body>
<h1>Welcome to My Blog</h1>
This is my first web blog.
Visit Google
</body>
</html>
Welcome to My Blog
This is my first web blog.
Visit Google

EX:32


SWISH
File
Edit
Examples
Help
204 users online
Search
Program
1 match(X, X).
2
3 check_match(X, Y) :-
4 match(X, Y),
5 write('Patterns match.').
6
7
8
check_match(cat, cat).
Patterns match.
true
?- check_match(cat, cat).

EX:33


SWISH
File
Edit
Examples
Help
212 users online
Search
Program
Program
vowel(a).
vowel(e).
vowel(i).
vowel(o).
vowel(u).
count_vowels([], 0).
count_vowels([H|T], Count) :-
vowel(H),
count_vowels(T, C),
Count is C + 1.
count_vowels([H|T], Count) :-
\+ vowel(H),
count_vowels(T, Count).
count_vowels([h,e,l,l,o], N).
N = 2
Next
10
100
1,000
Stop
?-
count_vowels([h,e,l,l,o], N).